Integral Einstein rigidity from polarized monopole classes

Abstract

We prove a scale-invariant, integral obstruction to Einstein metrics on closed oriented four-manifolds with suitable polarized Seiberg–Witten classes. The analytic statement uses actual solutions for every positive weight, but requires no Sobolev, injectivity-radius, diameter, volume, pointwise-curvature or curvature-derivative bound. Its proof retains the nonnegative errors in the monopole inequalities, bounds the spinor by a truncated weight, constructs a globally smooth polynomial Weyl model, and tests the contracted Bianchi identity against the divergence of that model. A bundle Hessian estimate proved below closes this self-test without a geometric compactness argument.

The resulting universal constant gives an integral Einstein gap for the connected sums occurring in LeBrun's Einstein obstruction. The class may depend on the metric, including its chamber when the positive Betti number is one; the same selected class must work for every weight in that metric's conformal class. Positive and negative Einstein constants and the separate Ricci-flat case are treated explicitly. The positive-budget application strengthens the numerical obstruction in LeBrun's Theorem 3.3 and Corollary 3.4. Negative topological budgets are handled directly by Chern–Gauss–Bonnet; the zero-budget, zero-surgery corner is not an Einstein nonexistence statement.

This manuscript makes the inherited integral-rigidity argument self-contained analytically. It does not assert priority for either that argument or the classical gauge-theoretic inputs.

1. Conventions and statements

Let M be a closed, connected, oriented smooth four-manifold, and let

$$V = \mathrm{Vol}_g(M), \qquad T = 2\chi(M) + 3\tau(M), \qquad P = \int_M |\mathrm{Ric}^\circ|^2.$$

Integrals and norms use g, unless stated otherwise. Ricci is a covariant two-tensor; W^+ is the trace-free symmetric curvature endomorphism of Λ^+, with its operator Hilbert–Schmidt norm. The Laplacian is nonnegative. Write $x_- := \max\{-x, 0\}$ and

$$4\pi^2 T = \int_M \left(\frac{s^2}{24} + 2|W^+|^2 - \frac{|\mathrm{Ric}^\circ|^2}{2} \right). \tag{1.1}$$

For a Spin^c structure with determinant line L, put

$$a = c_1(L) \in H^2(M;\mathbb{R}), \qquad a^+ = \mathrm{pr}_{H^+_g} a, \qquad p = (a^+)^2.$$

All cohomological squares mean intersection squares. The quadratic spinor map is normalized by

$$|\sigma(\Phi)|^2 = |\Phi|^4/8.$$

Definition 1.1 (the conformal monopole hypothesis). A pair (g, a) satisfies this hypothesis if $p > 0$ and one fixed Spin^c structure with first Chern class a has an unperturbed Seiberg–Witten

solution for every metric conformal to g. Equivalently for our purposes, for every smooth $f > 0$ there is a solution of

$$D_A\Phi = 0, \qquad -iF_A^+ = f\sigma(\Phi). \tag{1.2}$$

The implication from conformal solvability to (1.2) is proved in §2. The class is not allowed to change as f changes. A different class may be selected for a different initial metric.

Define the scale-invariant integral Einstein error by

$$\mathscr{E}(g) := \inf_{\lambda\neq 0} \frac{\int_M |\operatorname{Ric}_g - \lambda g|^2}{\lambda^2 V}. \tag{1.3}$$

Theorem 1.2 (integral rigidity). There is an absolute constant

$$\varepsilon_0 > 0 \tag{1.4}$$

with the following property. Suppose that for a metric g there is a class a satisfying Definition 1.1 and

$$(a^+)^2 \geqslant A > 0, \qquad a^2 \geqslant T. \tag{1.5}$$

Then

$$\boxed{\mathscr{E}(g) + \left(\frac{T}{A} - \frac{2}{3}\right)_+ \geqslant \varepsilon_0.} \tag{1.6}$$

In particular, the conclusion holds for every metric if such a class can be selected metricwise, with the same number A. No regularity of the class selection is needed. The constant is existential and universal; this paper does not assign it a numerical value.

The main analytic ingredient is a calibration gap. Let

$$\begin{aligned}
&\mathscr{O} = \operatorname{Orb}\operatorname{diag}(-1/6, 1/12, 1/12) \subset S^2_0(\mathbb{R}^3),\\
&F = \operatorname{dist}_{\mathrm{HS}}(W^+, \mathscr{O})^2, \qquad E = \operatorname{Ric} + g, \qquad e = s + 4,\\
&\alpha = \sqrt{2/3}|W^+|, \qquad q = -s + 3\alpha,\\
&\mathscr{C} = \int_M (|E|^2 + F), \qquad \Delta_+ = \int_M q_+^2 - 72\pi^2 p,\\
&\kappa = (\mathscr{C} + \Delta_+)/V, \qquad B_a = (-a^2)_+/V.
\end{aligned} \tag{1.7}$$

Here $\mathscr{O}$ is interpreted in each oriented orthonormal frame of Λ^+; the distance is frame independent.

Theorem 1.3 (analytic calibration gap). Under Definition 1.1,

$$\Delta_+ \geqslant 0, \qquad (1 + B_a)\kappa \geqslant c_{\mathrm{cal}} > 0, \tag{1.8}$$

where c_{cal} is absolute.

We prove this theorem first. Throughout the analytic proof, C denotes an absolute positive constant whose value can change from line to line. Explicit intermediate constants are retained where they clarify the topological transfer, not to advertise an optimized final constant.

2. Actual weighted solutions and the retained monopole identities

2.1. Why the weighted equation has solutions

Let $\widehat{g} = e^{2u} g = f^{-2} g$, so $u = -\log f$. Under the fiberwise unitary spinor identification β, the conformal formulas are

$$\begin{aligned} D_A^{\widehat{g}}(e^{-3u/2}\beta\Phi) &= e^{-5u/2}\beta D_A^g \Phi, \\ \sigma_{\widehat{g}}(\beta v) &= e^{2u}\sigma_g(v) \quad \text{as covariant two-forms.} \end{aligned} \tag{2.1}$$

For an ordinary solution $(A, \widehat{\Phi})$ for $\widehat{g}$, set

$$\Phi = f^{-3/2}\beta^{-1}\widehat{\Phi}.$$

Then $D_A^g \Phi = 0$, and

$$\sigma_g(\Phi) = f^{-1}\sigma_{\widehat{g}}(\widehat{\Phi}).$$

The Hodge star on two-forms is conformally invariant in dimension four, so the ordinary curvature equation becomes (1.2). In particular H_g^+ does not change during this operation. This is the conformal weighted monopole construction of LeBrun, Lemma 3.7; the displayed formulas specify the normalization used here.

Set

$$\rho = |\Phi|^2, \qquad \psi = 2\sqrt{2}\sigma(\Phi). \tag{2.2}$$

Thus ψ is a smooth self-dual two-form, $|\psi| = \rho$, and ρ itself is smooth, including at spinor zeros. The harmonic projection of the determinant curvature gives

$$\int_M f^2\rho^2 = 8\|F_A^+\|_2^2 \geqslant 32\pi^2 p. \tag{2.3}$$

Only the squared norm is used, so the equivalent opposite sign convention for the Chern representative does not change this formula.

2.2. Nonnegative errors

The spinor Weitzenböck equation and the self-dual Hodge formula are

$$4\nabla_A^*\nabla_A\Phi + (s + f\rho)\Phi = 0, \tag{2.4}$$

$$\|d\psi\|_2^2 + \|d^*\psi\|_2^2 = \int_M \Big(|\nabla\psi|^2 + \frac{s}{3}\rho^2 - 2\langle W^+\psi, \psi\rangle\Big). \tag{2.5}$$

The differential of the Hopf quadratic map in (2.2) satisfies

$$|d\psi_z(v)|^2 = 4|z|^2|v|^2 - 4\Big(\mathrm{Im}\langle z, v\rangle\Big)^2 \leqslant 4|z|^2|v|^2. \tag{2.6}$$

The same identity applies covariantly because the map is equivariant and parallel. Pairing (2.4) with $\rho\Phi$ and integrating yields

$$2\int_M |d\rho|^2 + 4\int_M \rho|\nabla_A\Phi|^2 + \int_M s\rho^2 + \int_M f\rho^3 = 0. \tag{2.7}$$

Let w be the smallest eigenvalue of W^+. For a trace-free symmetric three-by-three matrix, $w \geqslant -\alpha$. Define

$$\begin{aligned} D_{\mathrm{amp}} &= 2\int_M |d\rho|^2, \\ D_{\mathrm{quad}} &= \int_M (4\rho|\nabla_A\Phi|^2 - |\nabla\psi|^2), \\ D_H &= \|d\psi\|_2^2 + \|d^*\psi\|_2^2 = 2\|d\psi\|_2^2, \\ D_{\mathrm{eig}} &= 2\int_M (\langle W^+\psi, \psi\rangle + \alpha\rho^2), \\ D_0 &= D_{\mathrm{amp}} + D_{\mathrm{quad}} + D_H + D_{\mathrm{eig}}. \end{aligned} \tag{2.8}$$

Each term is nonnegative. Equations (2.5)–(2.7) give the exact identity

$$D_0 = \int_M \Big(\frac{2}{3}q\rho^2 - f\rho^3\Big). \tag{2.9}$$

For $v = 3\rho/2$, this is

$$\int_M (qv^2 - fv^3) = \frac{27}{8}D_0. \tag{2.10}$$

2.3. Signed spectral and scalar estimates

Put

$$b = 2s/3 + 2w, \qquad h = b_-.$$

Using w, rather than $-\alpha$, in (2.5)–(2.7) gives

$$\int_M f\rho^3 \leqslant \int_M h\rho^2. \tag{2.11}$$

Choose positive smooth functions $h_j \geqslant h$ converging uniformly to h, and use the actual solutions with $f_j = \sqrt{h_j}$. Set

$$C_j = \int_M f_j^2\rho^2, \quad Z_j = \int_M f_j\rho^3, \quad Q_j = \int_M f_j^4.$$

Equation (2.11) and Hölder imply

$$32\pi^2 p \leqslant C_j \leqslant Q_j^{1/3} Z_j^{2/3} \leqslant Q_j^{1/3} C_j^{2/3}.$$

Consequently $32\pi^2 p \leqslant Q_j$, and uniform approximation proves

$$\int_M (2s/3 + 2w)_-^2 \geqslant 32\pi^2 p, \qquad \int_M q_+^2 \geqslant 72\pi^2 p. \tag{2.12}$$

The second follows from $b_- \leqslant (2/3)q_+$. This argument passes to a limit of inequalities, not to a limiting spinor or connection.

For $f = 1$, pairing (2.4) with Φ gives

$$\int_M \rho^2 \leqslant \int_M s_-\rho \leqslant \|s_-\|_2 \|\rho\|_2.$$

Together with (2.3), this proves

$$\int_M s_-^2 \geqslant 32\pi^2 p. \tag{2.13}$$

In particular $p > 0$ excludes a scalar-flat metric. These are the classical monopole curvature estimates in the normalization needed here; the retained terms in (2.8) will supply their quantitative use.

3. A bounded spinor with small retained errors

The weights used in §2.3 need not bound the spinor uniformly. For the calibration argument we instead set

$$Q = \max\{q, 3\}, \qquad f_0 = \max\{\sqrt{Q}, s_-/4\}. \tag{3.1}$$

Choose smooth majorants $f_j \geqslant f_0$ with $f_j \to f_0$ uniformly, and choose an actual solution for each f_j. At a positive maximum of ρ, (2.4) implies $s + f_j\rho \leqslant 0$. Hence

$$0 \leqslant \rho \leqslant 4, \qquad 0 \leqslant v \leqslant 6. \tag{3.2}$$

All errors $o_j(1)$ below tend to zero for this fixed smooth metric and fixed class. They require only uniform approximation of f_0, not bounds on derivatives of f_j. No assertion of convergence of the solutions is made.

The two truncations cost only scalar error:

$$0 \leqslant Q^2 - q_+^2 \leqslant 9e^2, \qquad 0 \leqslant f_0^2 - Q \leqslant \frac{9}{16}e^2. \tag{3.3}$$

Indeed, the first difference vanishes unless $q < 3$, which implies $s > -3$ and $e > 1$. For the second, a nonzero difference implies $s_- > 4\sqrt{3} > 6$, and then $s_- \leqslant 3|s + 4|$.

Define ideal moments using Q, not the full truncated weight:

$$C_* := \int_M Qv^2, \qquad Z_* := \int_M \sqrt{Q}v^3, \qquad E_Q := \int_M Q^2, \qquad P_* := C_* - Z_*. \tag{3.4}$$

Equation (2.10), $Q \geqslant q$, and $f_j \geqslant \sqrt{Q}$ give

$$P_* \geqslant \frac{27}{8}D_0 \geqslant 0. \tag{3.5}$$

The actual curvature moment is

$$C_{\mathrm{act}} := \int_M f_j^2 v^2 = 18\|F_A^+\|_2^2 \geqslant 72\pi^2 p.$$

By (3.2)–(3.3),

$$0 \leqslant C_{\mathrm{act}} - C_* \leqslant \frac{81}{4}\int_M e^2 + o_j(1).$$

Hölder gives $C_* \leqslant E_Q^{1/3} Z_*^{2/3}$. Since $Z_* \leqslant C_*$, it follows that $C_* \leqslant E_Q$. Thus, with

$$S_* := \Delta_+ + \frac{117}{4}\int_M e^2,$$

we have

$$0 \leqslant E_Q - C_* \leqslant S_* + o_j(1). \tag{3.6}$$

For completeness, two elementary moment inequalities recover all the errors needed below. Write

$$\theta = (Z_*/E_Q)^{1/3}, \qquad H_* = E_Q^{1/3} Z_*^{2/3} - C_*.$$

Here $0 < \theta \leqslant 1$, $H_* \geqslant 0$, and direct algebra gives

$$E_Q - C_* = H_* + \frac{1+\theta}{\theta^2}(P_* + H_*) \geqslant 2P_*. \tag{3.7}$$

For the probability measure $d\nu = Q^2 d\mu / E_Q$, let $r = v/\sqrt{Q}$, $c = \int r^2 d\nu$, $z = \int r^3 d\nu$, and $m = \int r d\nu$. We have $z \leqslant c \leqslant z^{2/3}$, hence $c \leqslant 1$. Also $c^2 \leqslant mz \leqslant mc$, so $m \geqslant c$. Therefore

$$\int_M Q(v - \sqrt{Q})^2 = E_Q(c - 2m + 1) \leqslant E_Q - C_*. \tag{3.8}$$

Equations (3.5)–(3.8) prove

$$\begin{gathered} D_0 \leqslant \frac{4}{27} S_* + o_j(1), \\ \int_M |d\rho|^2, \ \int_M |d\psi|^2 \leqslant \frac{2}{27} S_* + o_j(1), \\ \int_M Q(v - \sqrt{Q})^2 \leqslant S_* + o_j(1). \end{gathered} \tag{3.9}$$

The point of the ideal moments is that these conclusions do not require any estimate of $\int f_j^4$, which could be unnecessarily large.

4. A globally smooth approximate Weyl tensor

The norm of an element of $\mathscr{O}$ is $1/(2\sqrt{6})$. Consequently

$$\left\| q - \frac{9}{2} \right\|_2^2 \leqslant 2\|e\|_2^2 + 12 \int_M F \leqslant 12\mathscr{E}. \tag{4.1}$$

Clipping at 3 moves q toward $9/2$, so the same bound holds with Q. Since $Q \geqslant 3$, (3.8) implies

$$\|\rho - \sqrt{2}\|_2^2 \leqslant \frac{8}{27} S_* + \frac{2}{27} \left\| Q - \frac{9}{2} \right\|_2^2 + o_j(1). \tag{4.2}$$

Here the first term follows by writing $\rho - \sqrt{2} = (2/3)(v - \sqrt{Q}) + (2/3)(\sqrt{Q} - \sqrt{9/2})$ and using $Q \geqslant 3$.

To estimate direction, choose a measurable rank-one orthogonal projection Π onto a lowest eigenspace of W^+. Then

$$A_0 = I/12 - \Pi/4$$

is a nearest point of $\mathscr{O}$. On the good set $F \leqslant 1/256$, comparison of ordered eigenvalues with A_0 gives

$$\lambda_2(W^+) - \lambda_1(W^+) \geqslant 1/8.$$

Putting $y = (I - \Pi)\psi$, it follows from the eigenvalue error in (2.8) that

$$\int_{\{F \leqslant 1/256\}} |y|^2 \leqslant \frac{16}{27} S_* + o_j(1).$$

The complementary set has volume at most $256 \int F$, while $|\psi|^2 \leqslant 16$. Thus

$$\int_M |y|^2 \leqslant \frac{16}{27} S_* + 4096 \int_M F + o_j(1). \tag{4.3}$$

No derivative of Π will be taken. Instead define the polynomial endomorphism

$$A_\psi = \frac{\rho^2}{24} I - \frac{1}{8} \psi \otimes \psi, \qquad R = W^+ - A_\psi. \tag{4.4}$$

It is globally smooth, trace-free, and defined even at spinor zeros. Its eigenvalues are $-\rho^2/12, \rho^2/24, \rho^2/24$, and $|A_\psi|^2 = \rho^4/96$. The rank-one identity

$$|\psi \otimes \psi - \rho^2 \Pi|^2 = 2\rho^2 |y|^2$$

gives the useful comparison

$$|A_\psi - A_0|^2 \leqslant \frac{\rho^2}{16} |y|^2 + \frac{(\rho^2 - 2)^2}{48}. \tag{4.5}$$

For example, (4.5) follows by inserting the intermediate matrix $\rho^2 A_0/2$ and applying $|X + Y|^2 \leqslant 2|X|^2 + 2|Y|^2$. Since $\rho \leqslant 4$ and $(4 + \sqrt{2})^2 < 36$, (4.1)–(4.5) imply

$$\begin{aligned} \|R\|_2^2 &\leqslant \frac{44}{27} S_* + 8194 \int_M F + \frac{1}{9} \left\| Q - \frac{9}{2} \right\|_2^2 + o_j(1) \\ &\leqslant 2\Delta_+ + 8500 \mathscr{E} + o_j(1). \end{aligned} \tag{4.6}$$

In the final line we used $\int e^2 \leqslant 4 \int |E|^2$. In particular, each of

$$\|e\|_2^2, \quad \|\rho - \sqrt{2}\|_2^2, \quad \|R\|_2^2, \quad \|d\rho\|_2^2, \quad \|d\psi\|_2^2$$

is bounded by $C\kappa V + o_j(1)$.

5. A Hessian estimate that needs no Sobolev constant

Lemma 5.1. Let u be a smooth section of a metric or Hermitian vector bundle with compatible connection over a closed Riemannian four-manifold. Let $\mathscr{F}$ be its curvature, with component norm $\sum_{i,j} |\mathscr{F}_{ij}|^2_{\mathrm{HS}}$, and set $K = \|u\|_\infty$. Then

$$\begin{aligned} \|\nabla u\|_4^2 &\leqslant 4K \|\nabla^2 u\|_2, \\ \|\nabla^2 u\|_2^2 &\leqslant 2\|\nabla^*\nabla u\|_2^2 + 32K^2 \|\operatorname{Ric}\|_2^2 + 33K^2 \|\mathscr{F}\|_2^2. \end{aligned} \tag{5.1}$$

Proof. Take real parts of Hermitian inner products when necessary. Write $v_i = \nabla_i u$, $H_{ij} = \nabla_i v_j$, and $L = \sum_i H_{ii} = -\nabla^*\nabla u$. Integrating the divergence of $X_i = |v|^2\langle u, v_i\rangle$ gives

$$\int |v|^4 \leqslant K \int |v|^2(2|H| + |L|) \leqslant K\|v\|_4^2(2\|H\|_2 + \|L\|_2).$$

Because $|L| \leqslant 2|H|$ in dimension four, this proves the first inequality, including the case $\|v\|_4 = 0$.

For the second, antisymmetry gives

$$H_{ij} - H_{ji} = \mathscr{F}_{ij} u, \qquad \|H\|_2^2 = \int \langle H_{ij}, H_{ji}\rangle + \frac{1}{2}\sum_{i,j} \|\mathscr{F}_{ij} u\|_2^2. \tag{5.2}$$

The commutator on the bundle-valued one-form v yields

$$\sum_i \nabla_i \nabla_j v_i = \nabla_j L + \operatorname{Ric}_{jk} v_k + \sum_i \mathscr{F}_{ij} v_i.$$

Integrating by parts in the first term of (5.2) and using the skew-adjointness of $\mathscr{F}_{ij}$, we obtain the exact identity

$$\|H\|_2^2 = \|L\|_2^2 - \int \operatorname{Ric}_{ij}\langle v_i, v_j\rangle + \sum_{i,j} \int \langle \mathscr{F}_{ij} v_j, v_i\rangle + \frac{1}{2}\sum_{i,j} \|\mathscr{F}_{ij} u\|_2^2. \tag{5.3}$$

In particular no derivative of curvature appears. Hölder and the first part bound the middle terms in absolute value by

$$4K(\|\operatorname{Ric}\|_2 + \|\mathscr{F}\|_2)\|H\|_2.$$

Use Young's inequality to absorb half of $\|H\|_2^2$, and then $(x+y)^2 \leqslant 2(x^2+y^2)$. This gives exactly the second inequality in (5.1). No Sobolev inequality or curvature smallness has been used. □

5.2. Applying the lemma to the bounded spinor

The calibration budget gives

$$\begin{aligned} \int |W^+|^2 &\leqslant 2\int F + V/12, \\ \int |\operatorname{Ric}|^2 &\leqslant 2\int |E|^2 + 8V, \\ \int s^2 &\leqslant 8\int |E|^2 + 32V. \end{aligned} \tag{5.4}$$

Equations (3.3) and (4.1) also imply

$$\int f_j^2 \leqslant C(V + \mathscr{C}) + o_j(1). \tag{5.5}$$

Indeed $Q \leqslant 9/2 + |Q - 9/2|$; Cauchy–Schwarz and $\sqrt{V\mathscr{C}} \leqslant (V + \mathscr{C})/2$ suffice.

For the determinant connection, Chern–Weil and (3.2) give

$$\|F_A\|_2^2 = 2\|F_A^+\|_2^2 - 4\pi^2 a^2 \leqslant 4\int f_j^2 + 4\pi^2(-a^2)_+. \tag{5.6}$$

The positive-spinor connection curvature $\mathscr{F}_A$ is a fixed linear representation of the Riemannian and determinant curvatures. In the block decomposition of the Riemannian curvature operator, its nontrivial spin action uses the Λ^+ output: it sees W^+, s, and the trace-free Ricci block, but not W^-. Consequently (5.4)–(5.6) imply

$$\int |\mathscr{F}_A|^2 \leqslant C\Big(V + \mathscr{C} + (-a^2)_+\Big) + o_j(1). \tag{5.7}$$

From (2.4) and $\rho \leqslant 4$,

$$\|\nabla_A^*\nabla_A\Phi\|_2^2 \leqslant \frac{1}{2}\int s^2 + 8\int f_j^2 \leqslant C(V + \mathscr{C}) + o_j(1). \tag{5.8}$$

Apply Lemma 5.1 with $\|\Phi\|_\infty \leqslant 2$. Since A_ψ is a parallel polynomial of degree four in Φ,

$$|\nabla^2 A_\psi| \leqslant C\Big(|\Phi|^3|\nabla_A^2\Phi| + |\Phi|^2|\nabla_A\Phi|^2\Big).$$

We conclude that

$$\int \Big(|\nabla_A^2\Phi|^2 + |\nabla_A\Phi|^4 + |\nabla^2 A_\psi|^2\Big) \leqslant C\Big(V + \mathscr{C} + (-a^2)_+\Big) + o_j(1). \tag{5.9}$$

This proves the required second-derivative budget without differentiating the weight and without an estimate of the anti-self-dual Weyl curvature.

6. A weak Bianchi self-test and its incompatible lower bound

6.1. Tensor normalization and the Bianchi upper bound

For an orthonormal basis ϵ_a of Λ^+, put $\omega_{\ a} = \sqrt{2}\epsilon_a$. Associate to a symmetric endomorphism A the four-tensor

$$W(A)_{ijkl} = \frac{1}{2}\sum_{a,b} A_{ab}(\omega_{\ a})_{ij}(\omega_{\ b})_{kl}. \tag{6.1}$$

Let

$$J_\psi = \operatorname{div} W(A_\psi),$$

using the full three-index norm $\sum_{j,k,l}|J_{jkl}|^2$. Changing the overall sign of divergence changes none of the following squared norms.

The contracted Bianchi identity expresses $\operatorname{div} W^+$ as a parallel algebraic contraction of $\nabla \operatorname{Ric}^\circ$ and ∇s. This follows by contracting the Riemannian Bianchi identity, subtracting the Schouten part, and projecting the final two-form pair onto Λ^+; that projection is parallel. Since $\nabla \operatorname{Ric} = \nabla E$ and $\nabla s = \nabla \operatorname{tr} E$, integration by parts and $W(A_\psi) = W^+ - W(R)$ show, for every smooth test tensor Z, that

$$\left|\int \langle J_\psi, Z\rangle\right| \leqslant C(\|E\|_2 + \|R\|_2)\|\nabla Z\|_2. \tag{6.2}$$

Set $Z = J_\psi$. It is smooth, and $\|\nabla J_\psi\|_2 \leqslant C\|\nabla^2 A_\psi\|_2$. Equations (4.6) and (5.9) therefore give

$$\frac{1}{V}\int |J_\psi|^2 \leqslant C\sqrt{\kappa}\sqrt{1+\kappa+B_a} + o_j(1). \tag{6.3}$$

This is a weak identity tested against a smooth tensor already supplied by the spinor. It asserts no pointwise bound on $\nabla \operatorname{Ric}$.

6.2. The exact divergence identity

Define

$$B_\psi = \frac{s}{3}\rho^2 - 2\langle W^+\psi, \psi\rangle, \qquad C_\psi = \langle \iota_{\nabla\rho}\psi, *d\psi\rangle.$$

We now derive the identity used to lower-bound the left side of (6.3). Pairing the self-dual Hodge–Weitzenböck equation with $\rho^2\psi$ and integrating by parts gives

$$\int \rho^2\left(|\nabla\psi|^2 + B_\psi - 2|d\psi|^2\right) = 4\int \rho C_\psi - 2\int \rho^2|d\rho|^2. \tag{6.4}$$

To see the mixed terms explicitly, use $d^*\psi = -*\,d\psi$, $d^*(\rho^2\psi) = \rho^2 d^*\psi - 2\rho\iota_{\nabla\rho}\psi$, and $d(\rho^2\psi) = \rho^2 d\psi + 2\rho d\rho\wedge\psi$. The two exterior-derivative mixed terms add to $4\rho C_\psi$; the rough-Laplacian mixed term is $2\rho^2|d\rho|^2$.

The required pointwise first-jet identities are

$$|\nabla A_\psi|^2 = \frac{\rho^2}{32}|\nabla\psi|^2 + \frac{\rho^2}{96}|d\rho|^2, \tag{6.5}$$

$$|\nabla A_\psi|^2 - |J_\psi|^2 = \frac{\rho^2}{64}(|\nabla\psi|^2 - 2|d\psi|^2) + \frac{\rho^2}{32}|d\rho|^2 - \frac{\rho}{16}C_\psi. \tag{6.6}$$

Here is a direct component verification that fixes all normalization factors. At a point with $\rho > 0$, choose a parallel orthonormal frame so that

$$\begin{aligned}
\epsilon_1 &= (dx^0 \wedge dx^1 + dx^2 \wedge dx^3)/\sqrt{2},\\
\epsilon_2 &= (dx^0 \wedge dx^2 + dx^3 \wedge dx^1)/\sqrt{2},\\
\epsilon_3 &= (dx^0 \wedge dx^3 + dx^1 \wedge dx^2)/\sqrt{2},\\
\psi &= \rho\epsilon_1, \qquad \nabla_m\psi = c_m\epsilon_1 + a_m\epsilon_2 + b_m\epsilon_3.
\end{aligned}$$

Then $c_m = \partial_m\rho$, and differentiating (4.4) gives

$$\nabla_m A_\psi = \begin{pmatrix} -\rho c_m/6 & -\rho a_m/8 & -\rho b_m/8 \\ -\rho a_m/8 & \rho c_m/12 & 0 \\ -\rho b_m/8 & 0 & \rho c_m/12 \end{pmatrix}. \tag{6.7}$$

The squared matrix norm gives (6.5). For (6.6), insert (6.7) in $J_{jkl} = \sum_{m,a,b}(\nabla_m A)_{ab}(\epsilon_a)_{mj}(\epsilon_b)_{kl}$, use $\sum_{k,l}(\epsilon_b)_{kl}(\epsilon_c)_{kl} = 2\delta_{bc}$, and compute $d\psi_{ijk} = \nabla_i\psi_{jk} + \nabla_j\psi_{ki} + \nabla_k\psi_{ij}$ in the displayed basis. Explicitly,

$$*d\psi = \frac{1}{\sqrt{2}}\begin{pmatrix} -a_2 - b_3 - c_1 \\ a_3 - b_2 + c_0 \\ a_0 + b_1 - c_3 \\ -a_1 + b_0 + c_2 \end{pmatrix}, \qquad \iota_{\nabla\rho}\psi = \frac{\rho}{\sqrt{2}}(-c_1, c_0, -c_3, c_2).$$

These expressions and (6.7) give (6.6) by summing squares and collecting the mixed terms in C_ψ. Thus (6.5)–(6.6) are algebraic identities in the twelve arbitrary numbers c_m, a_m, b_m, with no field equation imposed. At a spinor zero, both ψ and its first derivative vanish, and $d\rho = 0$; the identities extend there as well.

Combining (6.4)–(6.6) eliminates $\int \rho^2|\nabla\psi|^2$ and gives

$$\int |J_\psi|^2 = -\frac{1}{64}\int \rho^2 B_\psi + \frac{1}{16}\int \rho^2|d\psi|^2 - \frac{5}{96}\int \rho^2|d\rho|^2 + \frac{1}{8}\int \rho C_\psi. \tag{6.8}$$

6.3. The positive lower bound and contradiction

Because $A_\psi\psi = -\rho^2\psi/12$, the curvature term in (6.8) is

$$-\frac{\rho^2 B_\psi}{64} = -\frac{s\rho^4}{192} - \frac{\rho^6}{384} + \frac{\rho^2\langle R\psi,\psi\rangle}{32}. \tag{6.9}$$

At $s=-4$, $\rho=\sqrt{2}$, and $R=0$, it equals $1/16$. On the interval $0\leqslant\rho\leqslant 4$, the error in (6.9) is bounded by

$$C\big(|e| + |\rho-\sqrt{2}| + |R|\big).$$

Also $|C_\psi|\leqslant\rho|d\rho||d\psi|$. Discard the nonnegative $|d\psi|^2$ term in (6.8), apply (3.9), (4.2), and (4.6), and divide by V. Cauchy–Schwarz gives

$$\frac{1}{V}\int |J_\psi|^2 \geqslant \frac{1}{16} - C(\sqrt{\kappa}+\kappa) + o_j(1). \tag{6.10}$$

For each fixed metric let the smooth-weight approximation errors tend to zero in (6.3) and (6.10). The two numerical inequalities remain valid without choosing any limit of the spinors. For a sufficiently small absolute $\kappa_0\leqslant 1$, the lower bound is at least $1/32$ when $\kappa\leqslant\kappa_0$. The upper bound then forces

$$\kappa(1+\kappa+B_a)\geqslant c_1>0.$$

As $1+\kappa+B_a\leqslant 2(1+B_a)$ in this range, (1.8) follows. For $\kappa>\kappa_0$, it follows after decreasing the same universal constant. This proves Theorem 1.3. □

7. From the calibration gap to topological integral rigidity

We now prove Theorem 1.2, first at the negative normalization. Assume (1.5), and put

$$r = \frac{1}{V}\int |\operatorname{Ric}+g|^2, \qquad d = \left(\frac{T}{A}-\frac{2}{3}\right)_+.$$

Continue to use $\alpha=\sqrt{2/3}|W^+|$ and $w=\lambda_{\min}(W^+)$, and introduce

$$t = s+24\alpha = s+8\sqrt{6}|W^+|, \qquad b = 2s/3+2w, \qquad G = (2s/3-2\alpha)^2 - b_-^2. \tag{7.1}$$

Since $b\geqslant 2s/3-2\alpha$, we have $G\geqslant 0$, including where the scalar curvature is positive. Direct expansion gives

$$t^2+18G = 216\left(s^2/24+2|W^+|^2\right) - 18b_-^2.$$

Apply (1.1) and (2.12) to obtain

$$\int (t^2 + 18G) \leqslant 108P + 864\pi^2 Ad. \tag{7.2}$$

Here $p \geqslant A$ and $T - 2p/3 \leqslant T - 2A/3 \leqslant Ad$. Since $\int |E|^2 = P + \frac{1}{4}\int e^2$, (7.2) implies

$$\int (|E|^2 + e^2 + t^2 + G) \leqslant 109rV + 864\pi^2 Ad. \tag{7.3}$$

We record the pointwise orbit comparison, including its scalar-positive branch. Set $z = w + \alpha$; trace-freeness gives $0 \leqslant z \leqslant \alpha$, and minimizing over the orbit gives exactly

$$F = \frac{3}{2}(\alpha - 1/6)^2 + z/2, \qquad \alpha = (t - e + 4)/24. \tag{7.4}$$

If $s \leqslant -3$, then $b < 0$ and $G = z(-8s/3 + 8\alpha - 4z) \geqslant 8z$. Thus

$$F \leqslant (t^2 + e^2)/192 + G/16.$$

If $s > -3$, then $e > 1$. Use $z \leqslant \alpha$, $4 - e \leqslant 3e^2$, and $t \leqslant (t^2 + 1)/2$ in (7.4) to get

$$F \leqslant (t^2 + 5e^2)/64.$$

In both cases,

$$F \leqslant e^2 + t^2 + G. \tag{7.5}$$

The second completed square is

$$27\left(s^2/24 + 2|W^+|^2\right) - q^2 = t^2/8.$$

It and (1.1) yield

$$\Delta_+ \leqslant \frac{27}{2} rV + 108\pi^2 Ad.$$

Combining this with (7.3)–(7.5),

$$\kappa \leqslant \frac{245}{2} r + 972\pi^2 \frac{A}{V} d. \tag{7.6}$$

The scalar estimate (2.13) gives

$$32\pi^2 A \leqslant \int s^2 \leqslant 32V + 8rV, \qquad \frac{A}{V} \leqslant \frac{4 + r}{4\pi^2}. \tag{7.7}$$

For $r \leqslant 1$, therefore,

$$\kappa \leqslant 1215(r+d). \tag{7.8}$$

When $r \leqslant 1/4$, reverse triangle inequality in L^2 and (1.1) also imply

$$4\pi^2 T \geqslant \left[\frac{(4-2\sqrt{r})^2}{24} - \frac{r}{2}\right] V \geqslant V/4 > 0. \tag{7.9}$$

The square condition $a^2 \geqslant T$ then gives $B_a = 0$. Apply Theorem 1.3 and (7.8). If $r > 1/4$, no argument is needed. In either case

$$r + d \geqslant \varepsilon_0, \qquad \varepsilon_0 := \min\{1/4, c_{\mathrm{cal}}/1215\}. \tag{7.10}$$

For reference, a weaker square hypothesis has a correspondingly nonuniversal version. If $a^2 \geqslant -C_A A$, $C_A \geqslant 0$, the same proof gives

$$r + d \geqslant \min\left\{1, \frac{c_{\mathrm{cal}}}{1215(1 + 5C_A/(4\pi^2))}\right\}. \tag{7.11}$$

If instead $a^2 \geqslant T - C_A A$, replace the first entry of that minimum by $1/4$, using (7.9). An arbitrary monopole class must not silently be assumed to have $a^2 \geqslant T$.

8. Positive scale, arbitrary scale, and the Ricci-flat case

8.1. The positive normalization

Put $r_+ = V^{-1} \int |\operatorname{Ric} - g|^2$. Since $s_-^2 \leqslant (s-4)^2 \leqslant 4|\operatorname{Ric} - g|^2$, (2.13) gives

$$A/V \leqslant r_+/(8\pi^2).$$

If $r_+ \leqslant 1/4$, the analogue of (7.9) gives $4\pi^2 T/V \geqslant 1/4$. If also $d \leqslant 1/4$, then $T/A \leqslant 2/3 + d$, so

$$\frac{1}{4} \leqslant \frac{4\pi^2 T}{V} \leqslant \frac{r_+}{2}\left(\frac{2}{3} + d\right) \leqslant \frac{11}{96},$$

a contradiction. In particular $r_+ + d > 1/4$.

8.2. Scale optimization

For $\lambda \neq 0$, apply the appropriate normalized result to $|\lambda| g$. The positive harmonic subspace and the conformal solvability hypothesis are unchanged. The normalized error becomes

$$\frac{\int |\operatorname{Ric}_g - \lambda g|^2}{\lambda^2 V}.$$

Taking the infimum proves (1.6). This completes Theorem 1.2. □

For interpretation, write

$$R_2 = \int |\operatorname{Ric}|^2, \qquad S = \int s, \qquad L_s = \int (s - S/V)^2.$$

When $R_2 > 0$, direct minimization in $1/\lambda$ gives

$$\mathscr{E}(g) = 4 - \frac{S^2}{VR_2} = \frac{4P + L_s}{R_2}. \tag{8.1}$$

This formula includes $S = 0$, when the infimum is 4 and need not be attained. If $R_2 = 0$, definition (1.3) gives $\mathscr{E}(g) = 4$, not zero. Thus a lower bound for $\mathscr{E}$ alone does not exclude a Ricci-flat metric. Ricci-flat metrics in Theorem 1.2 are excluded separately by (2.13), since $p \geqslant A > 0$.

For an Einstein metric with nonzero Einstein constant, $\mathscr{E} = 0$. A pointwise operator error also dominates the integral one: for $\sigma = \pm 1$,

$$\delta_\sigma(g) := \| \operatorname{Ric}^\sharp - \sigma I \|_{L^\infty,\mathrm{op}} \quad \Longrightarrow \quad \mathscr{E}(g) \leqslant 4\delta_\sigma(g)^2. \tag{8.2}$$

There are exactly four Ricci eigenvalues, which explains this factor.

9. Metricwise polarized classes and chamber bookkeeping

The analytic theorem needs the quantifiers

$$\boxed{\begin{array}{c}\text{for every } g \text{ there exists a Spin}^c \text{ structure } \mathfrak{s}_g \\ \text{such that for every } f > 0, \text{ equation (1.2)} \\ \text{has a solution for the same } \mathfrak{s}_g.\end{array}} \tag{9.1}$$

It does not require one universal choice of signs for all metrics. This distinction is essential when $b_+ = 1$.

9.1. The connected-sum input with insertions

Let X have $b_+(X) > 0$, and suppose a Spin^c structure $\mathfrak{s}$, with real first Chern class c, has a nonzero Seiberg–Witten invariant evaluated on a homogeneous insertion z. The invariant is defined on

$$\mathbb{A}(X) = \mathbb{Z}[U] \otimes \Lambda^* H_1(X;\mathbb{Z})/\text{torsion}, \qquad \deg U = 2, \quad \deg H_1 = 1.$$

Its expected dimension is $(c^2 - T_X)/4$, where $T_X = 2\chi(X) + 3\tau(X)$. Assume

$$T_X > 0, \qquad A := c^2 \geqslant T_X. \tag{9.2}$$

The second inequality follows automatically from the nonnegative degree of a nonzero ordinary invariant, but is displayed because it is used in the proof. For $b_+ = 1$, the nonzero invariant here must be in the small-perturbation chamber at the relevant polarization, not merely in some unspecified chamber.

Let

$$M = X\#k\overline{\mathbb{CP}}^2\#\ell(S^1\times S^3), \qquad D = k+4\ell, \qquad T_M = T_X - D. \tag{9.3}$$

For a prescribed positive polarization $H = H^+_g$, choose the signs of the exceptional generators E_j so that

$$c^+_H \cdot E_j \leqslant 0,$$

and set

$$a_H = c - \sum_{j=1}^{k} E_j. \tag{9.4}$$

Then

$$\begin{aligned} (a^+_H)^2 &= (c^+_H)^2 - 2\sum_j c^+_H\cdot E_j + \left|\sum_j (E_j)^+_H\right|^2 \geqslant (c^+_H)^2 \geqslant c^2 = A, \\ a^2_H &= A - k \geqslant T_X - k - 4\ell = T_M. \end{aligned} \tag{9.5}$$

The ordinary blow-up formula with exceptional coefficient ± 1 preserves expected dimension and preserves the original insertion z. The $S^1\times S^3$ operation raises expected dimension by one and adds the degree-one insertion of its new circle. After ℓ operations the nonzero invariant is therefore evaluated on

$$z\,\mu(C_1)\dots\mu(C_\ell). \tag{9.6}$$

These statements include positive-dimensional base invariants. They are the special cases of the blow-up and surgery formulas in Ozsváth–Szabó, Theorem 3.2, Proposition 2.2 and Remark 2.3. The last remark supplies the chamber-compatible version when $b_+ = 1$. Using only a zero-dimensional numerical invariant and omitting the insertions in (9.6) would not justify the general statement.

9.2. What the chamber condition means

For $b_+(X) = 1$, choose a unit forward generator ω_H of the positive line on M so that $c\cdot\omega_H > 0$. The X component ω_X has positive square, lies in the corresponding forward cone, and satisfies $c\cdot\omega_X > 0$, because $c^2 > 0$. The chosen exceptional signs give $E_j\cdot\omega_H \leqslant 0$, hence

$$a_H\cdot\omega_H > 0. \tag{9.7}$$

Thus the small-perturbation discriminants for the base and the selected blow-up structure have the same sign in corresponding chamber conventions. Adding $S^1\times S^3$ leaves the second-cohomology polarization unchanged. The gluing formulas consequently give a nonzero invariant in the chamber

containing zero perturbation for this selected class, provided that was the specified nonzero base chamber.

For any metric with this polarization, take generic perturbations tending to zero in that chamber. Nonzero invariant gives solutions, and the usual fixed-metric Seiberg–Witten compactness gives an unperturbed solution. A reducible limit would have $F_A^+ = 0$, contradicting $(a_H^+)^2 \geqslant A > 0$. Every metric conformal to g has the same polarization and the same nonzero discriminant sign. Thus the *same* class (9.4) satisfies Definition 1.1 for every weight, as required by (9.1).

For $b_+ > 1$, there is no chamber ambiguity in this argument. For $b_+ = 1$, the statement must not be replaced by the assertion that every exceptional sign choice is an ordinary all-metric monopole class.

9.3. The minimal general-type case

If X is a minimal compact complex surface of general type with its complex orientation, then

$$A = c_1(X)^2 = T_X > 0.$$

The canonical Spin^c structure has a nonzero Seiberg–Witten invariant. For $b_+ > 1$, the standard Kähler/symplectic nonvanishing statement supplies the input directly. For $b_+ = 1$, the canonical class K has $K^2 > 0$ and positive pairing with a Kähler form. It is therefore in the forward time-like cone and pairs positively with every forward period ray. For $c = -K$, the small-perturbation chamber agrees with the canonical nonzero chamber, up to the simultaneous choice of sign convention for the chamber discriminant. This is the light-cone/chamber argument underlying LeBrun's polarized construction, Proposition 3.2. The chamber convention and canonical nonvanishing can also be read in Li–Liu, §2.1 and Theorem 2.2.

The positivity of K on the symplectic or Kähler form matters: $b_+ = 1$ and $K^2 > 0$ alone do not identify the nonzero small-perturbation chamber for an arbitrary symplectic manifold.

10. The Einstein obstruction and its precise relation to LeBrun

Theorem 10.1 (connected-sum integral obstruction). Under the gauge-theoretic and chamber assumptions of §9.1, every metric on (9.3) satisfies

$$\mathscr{E}(g) + \left(\frac{T_M}{A} - \frac{2}{3}\right)_+ \geqslant \varepsilon_0. \tag{10.1}$$

It also satisfies the slightly weaker but base-topological statement

$$\boxed{\mathscr{E}(g) + \left(\frac{1}{3} - \frac{D}{T_X}\right)_+ \geqslant \varepsilon_0.} \tag{10.2}$$

For minimal general type, $A = T_X = c_1(X)^2$, so the two statements coincide.

Proof. Equation (9.5) and the conformal solvability just proved give all the hypotheses of Theorem 1.2. If $T_M > 0$, then $A \geqslant T_X$ implies $T_M/A \leqslant T_M/T_X$, yielding (10.2). If $T_M \leqslant 0$, both positive-part

terms are zero. This case split avoids reversing an inequality by dividing a negative numerator by the larger denominator. □

In particular, put $\gamma = \varepsilon_0/2$. Then

$$D \geqslant (1/3 - \gamma)T_X \quad \Longrightarrow \quad \mathscr{E}(g) \geqslant \varepsilon_0/2 \quad \text{for every metric } g. \tag{10.3}$$

There is no Einstein metric in this band: nonzero Einstein constants contradict (10.3), and zero Einstein constant contradicts (2.13). At either normalized sign, (8.2) also gives

$$\delta_\sigma(g) \geqslant \sqrt{\varepsilon_0/8}. \tag{10.4}$$

An Einstein metric would in fact require $D/T_X \leqslant 1/3 - \varepsilon_0$. The equality endpoint of that necessary condition is not excluded by (10.2) alone; the closed band (10.3) avoids that endpoint issue.

10.2. Exact source scope

The version of LeBrun's paper verified here is arXiv:math/0003068v1, March 2000. Its Proposition 3.2 constructs the metricwise polarized classes on the connected sums in (9.3). Its Theorem 3.3 states the classical threshold

$$k + 4\ell \geqslant T_X/3,$$

and Corollary 3.4 specializes the result to blow-ups of minimal complex surfaces of general type. We use the ordinary nontrivial Seiberg–Witten input, including its insertion and chamber meaning, not merely the topology of a manifold with the same Euler characteristic and signature. The proof above handles the positive-budget case of the general theorem, not only its general-type corollary.

The numerical band below the classical threshold is nonvacuous in families with unbounded budget. For example, let $X_m = \Sigma_2 \times \Sigma_m$, $m \geqslant 2$. Then $A = T_{X_m} = 8(m-1)$. With $\ell = 0$ and $k = \lceil A/3 \rceil - 1$,

$$0 < 1/3 - k/A \leqslant 1/A \longrightarrow 0.$$

For sufficiently large m, these ratios lie in (10.3). Since the universal analytic constant has not been made numerical, this argument does not specify the first qualifying m.

10.3. Nonpositive budgets and the exceptional zero corner

There is a direct all-metric formula, independent of monopoles. For $R_2 > 0$, (1.1) and (8.1) give

$$\mathscr{E}(g) = 1 + \frac{12\int |W^+|^2 - 24\pi^2 T + L_s}{R_2}. \tag{10.5}$$

Thus $T \leqslant 0$ implies $\mathscr{E}(g) \geqslant 1$. For a Ricci-flat metric its value is 4 by definition, so the same lower bound holds. This lower bound is not a Ricci-flat exclusion.

For an Einstein metric, (1.1) reduces to

$$4\pi^2 T = \int (s^2/24 + 2|W^+|^2) \geqslant 0.$$

Consequently, when $T_X < 0$, or when $T_X = 0$ and $D > 0$, the connected sum has $T_M < 0$ and admits no Einstein metric. These are classical Chern–Gauss–Bonnet consequences and require no positive-budget monopole argument.

When $T_X = D = 0$, however, equality permits Ricci-flat, anti-self-dual metrics in the chosen orientation. K3 and the four-torus provide the standard zero-budget examples. The unqualified statement of Theorem 3.3 in the verified preprint includes this corner, but its appeal to the Hitchin–Thorpe inequality does not exclude these equality cases. We therefore do **not** claim an unqualified subsumption of that literal zero-corner wording, nor infer an exception-free statement for a separately unverified published version. The conclusions established here are: the positive-budget integral strengthening with the stated gauge input; the classical negative-T_M exclusion; and an explicit reservation of the Ricci-flat zero corner.

11. Logical dependencies and limitations

The analytic proof has the following complete chain:

$$\begin{aligned}
&\text{one actual conformal Spin}^c \text{ structure} \implies \text{solutions for smooth truncated weights} \\
&\implies \rho \leqslant 4 \text{ and small retained first-order defects} \implies W^+ \approx A_\psi \text{ with } A_\psi \text{ smooth} \\
&\implies \text{bundle Hessian budget and weak Bianchi upper bound} \\
&\implies \text{incompatibility with the exact positive divergence term.}
\end{aligned}$$

The proof neither differentiates an eigenprojection nor normalizes ψ across its zero set. Smooth approximation of weights is used only for numerical inequalities at a fixed metric; there is no claimed uniform regularity or limiting solution for a nonsmooth weight. The Hessian estimate uses the positive-spinor connection, which is why no L^2 budget for W^- enters. The topological theorem additionally needs the class-square bound and the actual class/chamber construction; the scalar curvature inequality by itself is not a substitute for those inputs.

This argument proves an obstruction and an integral gap. It does not prove convergence of near-Einstein metrics, classify all equality metrics of other curvature functionals, produce an Einstein metric, or settle arbitrary pointwise Ricci-pinching problems without the monopole/topological hypotheses displayed above.

References and provenance

- Claude LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, arXiv:math/0003068v1, especially Proposition 3.2, Theorem 3.3 and Corollary 3.4. The source version and the zero-budget caveat are specified in §10.
- Peter Ozsváth and Zoltán Szabó, *Higher Type Adjunction Inequalities in Seiberg–Witten Theory*, arXiv:math/0005268, especially Proposition 2.2, Remark 2.3 and Theorem 3.2. These supply insertion-aware surgery and blow-up formulas.
- Tian-Jun Li and Ai-Ko Liu, *Uniqueness of Symplectic Canonical Class, Surface Cone and Symplectic Cone of 4-Manifolds with $b^+ = 1$*, arXiv:math/0012048, §2.1 and Theorem 2.2, for chamber conventions and canonical nonvanishing.

- Claude LeBrun, *Four-Manifolds, Curvature Bounds, and Convex Geometry*, arXiv preprint, especially Lemma 3.7, for the conformal weighted-equation input.

This text consolidates the local weighted-monopole, integral-calibration, all-sign, chamber-selection and LeBrun-comparison arguments. Those earlier working records are provenance, not proof dependencies of the analytic argument presented here. Numerical and symbolic checks are useful for detecting transcription errors; none is used in place of a mathematical identity or a gauge-theoretic existence theorem.